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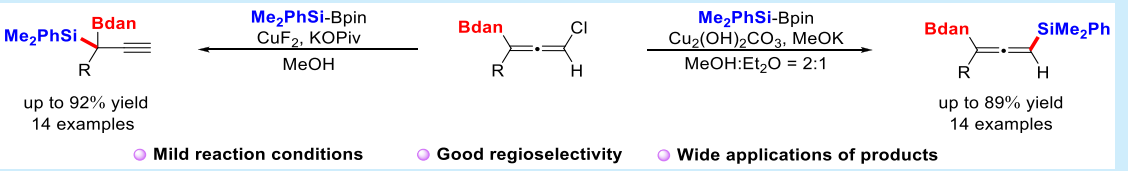
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Cu-Catalyzed Regioselective Silylation of Chloro-Substituted Allenyl-Bdan

Qi Ran, Ke-Fan Wu, and Yun-He Xu*

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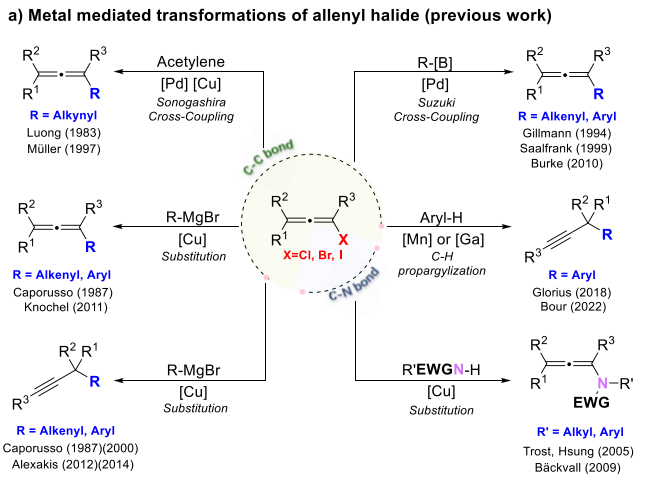


ABSTRACT: A copper-catalyzed efficient regioselective silylation reaction of chloro-substituted allenyl-Bdan was developed. Under mild reaction conditions, allenyl and propargyl silane compounds can be selectively obtained in moderate to high yields by adjusting the bases and solvents used in the reactions. This study offers direct and efficient methods for synthesizing multifunctionalized allenyl and propargyl silane compounds from the same initial material of chloro-substituted allenyl-Bdan.

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Allenes are recognized as stable organic compounds with two cumulated carbon−carbon double bonds, featuring two π orbitals perpendicular to each other (Scheme 1a).[1] This unique structure, coupled with their substituent-loading capability, gives them distinctive reactivity and significant potential in organic synthesis. Among them, allenyl halides serve as valuable building blocks in chemical synthesis as a result of their ease of conversion into various functional groups. Most examples discovered to date focus on constructing carbon−carbon bonds. For instance, Sonogashira cross-coupling and Suzuki cross-coupling of allenyl halide compounds have been previously reported.[2] Recently, transition-metal-catalyzed C−H propargylation has emerged as another method. The use of bromoallenes as coupling partners in direct C−H propargylation highlights their unique potential as building blocks in organic synthesis.[3] Additionally, copper-catalyzed regioselectivity substitution of allenyl halides has been extensively documented.[4] In 1987, Caporusso et al. reported the first copper-catalyzed alkylation of bromoallenes for alkyne and allene preparation, albeit yielding a mixture of both products. In 2011, Knochel and co-workers[4d] used choroallenes to synthesize trisubstituted allene compounds. Subsequently, Alexakis and co-workers[4e,f] selectively produced trisubstituted allenes and tetrasubstituted propargyl compounds from choroallenes using different solvents. Furthermore, C−N bond formation has been achieved. In 2005, Trost and Stiles[4i] documented copper-catalyzed coupling of allenyl halides with amides, carbamates, and ureas to synthesize allenamides. Hsung's group[4j] established a series of chiral allenamides through stereospecific amidation of optically enriched allenyl iodides. Despite the versatile utilities of allenamides, very few other C−X (e.g., C−Si) bond trans-

Scheme 1. Transformations of Allenyl Halides



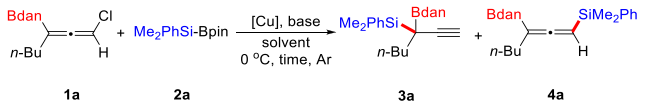
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Table 1. Optimization of Reaction Conditions[a,b,c]



1a 2a 3a 4a

entry	catalyst (mol %)	base (equiv)	solvent	3a/4a	yield (%)
1	CuF_2 (10)	Et_3N (2.0)	MeOH	82:18	56
2	CuF_2 (10)	CsOAc (2.0)	MeOH	>99:0	86
3	CuF_2 (10)	KOPiv (2.0)	MeOH	>99:0	95
4	CuF_2 (10)	Cs_2CO_3 (2.0)	MeOH	26:74	77
5	$CuCl_2$ (10)	Cs_2CO_3 (2.0)	MeOH	34:66	80
6	$Cu(acac)_2$ (10)	Cs_2CO_3 (2.0)	MeOH	45:55	64
7	$Cu_2(OH)_2CO_3$ (5)	Cs_2CO_3 (2.0)	MeOH	25:75	86
8	$Cu_2(OH)_2CO_3$ (5)	KOH (2.0)	MeOH	20:80	71
9	$Cu_2(OH)_2CO_3$ (5)	NaOEt (2.0)	MeOH	25:75	84
10	$Cu_2(OH)_2CO_3$ (5)	KOMe (2.0)	MeOH	17:83	83
11	$Cu_2(OH)_2CO_3$ (5)	KOMe (2.0)	MeOH/THF = 2:1	0:>99	47
12	$Cu_2(OH)_2CO_3$ (5)	KOMe (2.0)	MeOH/DME = 2:1	0:>99	35
13	$Cu_2(OH)_2CO_3$ (5)	KOMe (2.0)	MeOH/Et_2O = 2:1	0:>99	71
14[c]	$Cu_2(OH)_2CO_3$ (5)	KOMe (2.2)	MeOH/Et_2O = 2:1	0:>99	80
15[d]	CuF_2 (5)	KOPiv (2.0)	MeOH	>99:0	95 (92[e])
16[c,d]	$Cu_2(OH)_2CO_3$ (2.5)	KOMe (2.2)	MeOH/Et_2O = 2:1	0:>99	82 (82[e])

[a]Reaction conditions: compound **1a** (0.2 mmol), compound **2** (0.4 mmol), base (0.4 mmol), and catalyst in solvent at 0 °C for 12 h. [b]The yield was determined by ^1H nuclear magnetic resonance (NMR) using 1,1,2,2-tetrachloroethane as an internal standard. [c]Compound **2a** (0.44 mmol). [d]The reaction time is 4 h. [e]Isolated yield.

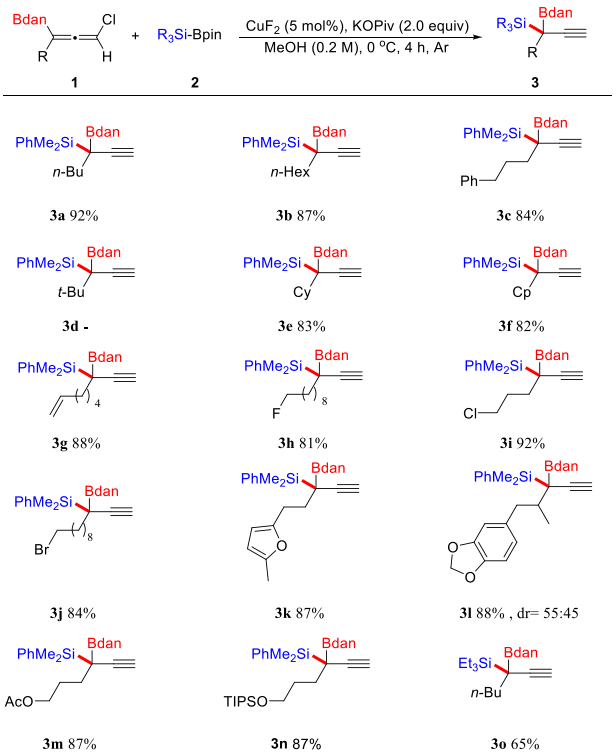


Figure 1. Substrate scope for synthesis of product **3**. (a) Reaction conditions: compound **1** (0.2 mmol), compound **2** (0.4 mmol), KOPiv (0.4 mmol), and CuF_2 (5 mol %) in MeOH (1.0 mL) at 0 °C for 4 h. (b) Isolated yield.

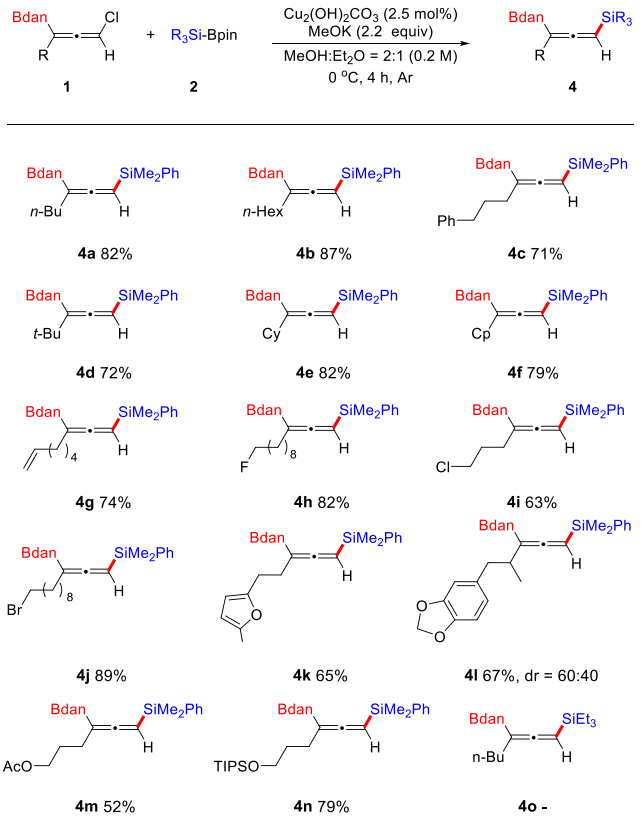
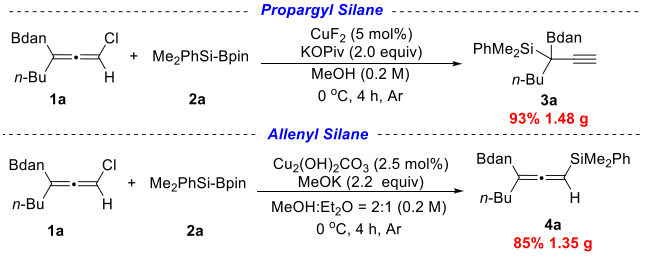


Figure 2. Substrate scope for synthesis of product **4**. (a) Reaction conditions: compound **1** (0.2 mmol), compound **2** (0.44 mmol), KOMe (0.44 mmol), and $Cu_2(OH)_2CO_3$ (2.5 mol %) in MeOH/Et_2O = 2:1 (1.0 mL) at 0 °C for 4 h. (b) Isolated yield.

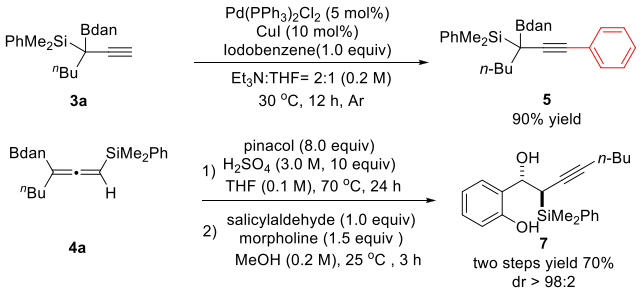
formations have been detailed. Meanwhile, derivatization of carbon-substituted allenyl and propargyl compounds without functional groups has been unsatisfactory. Hence, there is a strong imperative to synthesize more active allenyl and propargyl compounds.[5] Motivated by our ongoing enthusiasm for developing multifunctionalized allenyl and propargyl silane compounds, we present copper-catalyzed silylation reactions of chloro-substituted allenyl-Bdan to synthesize these versatile

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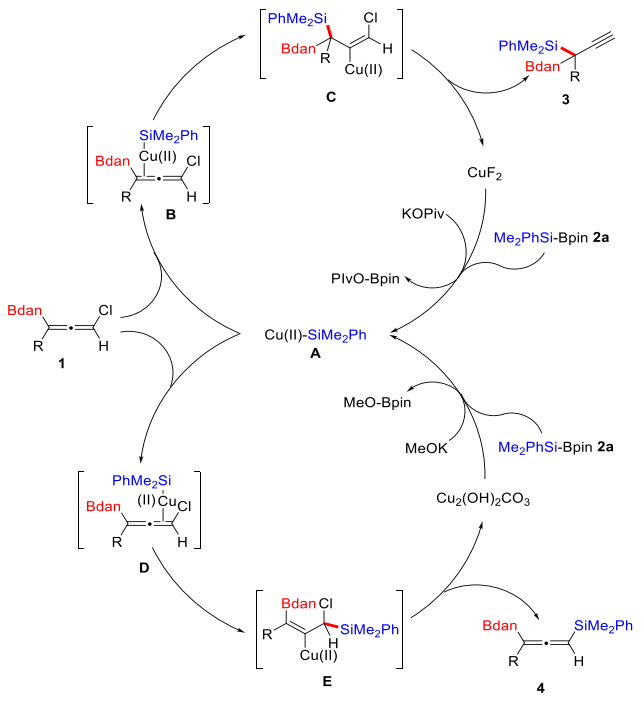
Scheme 2. Gram-Scale Synthesis of Products 3a and 4a



Scheme 3. Transformation of Products 3a and 4a



Scheme 4. Proposed Mechanistic Pathway



compounds (Scheme 1b). By adjusting the choice of solvents and bases in the reactions, we can accurately predict and control the product distributions.

When compound **1a** underwent a reaction with 2.0 equiv of pinB-SiMe$_2$Ph (**2a**) in the presence of Et$_3$N and a catalytic amount of CuF$_2$, propargyl silane **3a** and allenylsilane **4a** were produced in an 82:18 ratio with a yield of 56% (see entry 1 in Table 1). Subsequently, various bases were tested (Table S1 of the Supporting Information). By replacement of KOPiv with Et$_3$N, the yield of product **3a** increased to 95% (entry 3 in Table 1). Employing Cs$_2$CO$_3$ instead of Et$_3$N resulted in a shift toward the formation of product **4a** in the **3a**/**4a** selectivity (entry 4 in Table 1). Further optimization aimed to enhance the yield of product **4a** through different catalysts (Table S2 of the Supporting Information). It was observed that Cu$_2$(OH)$_2$CO$_3$ enhanced the yield to 86% with moderate regioselectivity (**3a**/**4a** = 25:75; entry 7 in Table 1). Replacing Cs$_2$CO$_3$ with MeOK improved the selectivity of products **3a** and **4a** to 17:83 while achieving an 83% yield (entry 10 in Table 1). Subsequently, a variety of solvents were assessed (entries 11−13 in Table 1), with the addition of ethyl ether significantly enhancing selectivity but slightly reducing the yield (entry 13 in Table 1). Remarkably, increasing the amounts of compound **2a** and KOMe used in the reaction to 2.2 equiv led to an excellent yield and single selectivity for the formation of product **4a** (entry 14 in Table 1). The reaction proceeded without a notable decrease in the yield when shortening the time and reducing the catalyst (entries 15 and 16 in Table 1).

With the optimized conditions established, we proceeded to explore the range of reactions for generating product **3**. Substrates with diverse alkyl groups were efficiently transformed into respective Bdan-substituted propargylic silanes, yielding high amounts of products with remarkable selectivity (**3a**−**3c**; Figure 1). We were pleased to discover that the secondary alkyl substituents exhibited compatibility with the reaction environment (**3e** and **3f**; Figure 1). However, the desired product could not be synthesized when utilizing *t*-butyl substituents (**3d**; Figure 1). Additionally, a variety of primary alkyl-substituted substrates containing potentially reactive functional groups were also accommodated (**3g**−**3j**; Figure 1), offering opportunities for subsequent derivatization. Notably, compounds featuring allenyl-Bdan with reactive moieties, like furan (**3k**), acetal (**3l**), ester (**3m**), and silyl ether (**3n**), demonstrated compatibility with the reaction conditions. When pinB-SiMe$_2$Ph replaced pinB-SiEt$_3$, product **3o** was obtained at a yield of 65% with single selectivity.

Following this, we initiated an exploration into the reaction scope for synthesizing product **4**. The reaction of different alkyl-substituted allenyl-Bdan proceeded smoothly, resulting in the formation of the respective allenylsilanes with high yields (**4a**−**4c**, as depicted in Figure 2). Moreover, when secondary alkyl or tertiary alkyl-substituted allenyl-Bdan compounds were utilized, the corresponding products were also obtained in significant yields (**4d**−**4f**; Figure 2). Various functional groups, including double bonds (**4g**), diverse halides (**4h**−**4j**), furan (**4k**), acetal (**4l**), ester (**4m**), and silyl ether (**4n**) present in the substrates, were well-accommodated. Regrettably, the substitution of pinB-SiMe$_2$Ph with pinB-SiEt$_3$ prevented the synthesis of product **4o**.

To showcase the practical applicability of the products, we conducted a series of transformative reactions involving products **3a** and **4a**. Initially, a gram-scale reaction of compound **1a** was performed on a 4 mmol scale, resulting in a 93% yield for product **3a** (1.48 g) and an 85% yield for product **4a** (1.35 g) (see Scheme 2). Subsequently, product **3a** was employed for C−C bond formation reactions, specifically Sonogashira coupling reactions using iodobenzene, yielding product **5** at a 90% yield.[4f] Subsequently, product **4a** was combined with pinacol and sulfuric acid in tetrahydrofuran. Without further purification, crude product **6** directly proceed to the next step of the reaction, which mixed with salicylaldehyde in the presence of morpholine, leading to the isolation of corresponding homopropargylic alcohol **7** with a yield of 70% (Scheme 3).[5b]

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According to prior reports and our experimental findings,[4d,6] we have put forward a feasible mechanistic pathway (Scheme 4). Initially, in methanol, the combination of pinB-$SiMe_2Ph$, a copper catalyst, and a base can form species **A**. Subsequently, π complex **B** or **D** may arise from the interaction of species **A** with allenyl-Bdan. Following this, an antiselective elimination process involving π complex **B** or **D** takes place, leading to the creation of copper(II) intermediate **C** or **E**. β-Elimination from intermediate **C** or **E** then yields the desired product **3** or **4**, with the copper catalyst being liberated for the subsequent catalytic cycle.

In summary, we have devised an innovative copper-catalyzed technique for producing multifunctionalized allenyl and propargyl silane compounds from a common initial material of chloro-substituted allenyl-Bdan. The resulting allenyl and propargyl silane compounds serve as versatile reagents capable of undergoing diverse conversions to generate valuable synthetic products.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c00864.

Experimental details, characterization data, and spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Yun-He Xu − *Department of Chemistry, University of Science and Technology of China, Hefei, Anhui 230026, People's Republic of China;* orcid.org/0000-0001-8817-0626; Email: xyh0709@ustc.edu.cn

Authors

Qi Ran − *Department of Chemistry, University of Science and Technology of China, Hefei, Anhui 230026, People's Republic of China*

Ke-Fan Wu − *Department of Chemistry, University of Science and Technology of China, Hefei, Anhui 230026, People's Republic of China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.4c00864

Notes

The authors declare no competing financial interest.

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